# Regression Discontinuity Design

Lecture 4: Exploiting cutoffs to identify causal effects

# Where We Are

# Recap: our toolkit so far

We've been building a toolkit for causal inference. Each method handles selection bias differently.

- RCTs / A-B testing: eliminate selection bias by randomizing
- Difference-in-differences: compare *changes* across groups
- Panel methods (FE, FD): use within-unit variation over time

Today: a fourth approach. Use a sharp cutoff rule — where being just above or just below the threshold is essentially random — to identify causal effects.

# Today's roadmap

1. What problem does RDD solve?
2. How does RDD work? (visually *and* via regression)
3. When does RDD fail?
4. Sharp vs. fuzzy discontinuities
5. Identifying assumptions and how to defend them
6. Where RDD is used in practice

**Running example:** The National Merit Scholarship Program. Students who score above a cutoff on the PSAT receive a scholarship. We want to know whether this scholarship affects later outcomes — say, college GPA.

# What Problem Does RDD Solve?

# The setup: National Merit Scholarship

The National Merit Scholarship Program awards scholarships based on PSAT scores.

Rule (simplified):

- Score $\geq$ cutoff $\Rightarrow$ receive scholarship
- Score $<$ cutoff $\Rightarrow$ no scholarship

**Question:** Does receiving the scholarship affect college GPA?

The scholarship comes with prestige, recognition, financial support — all of which might affect academic outcomes. But is the effect real, or are we just observing that high-scoring students were going to do well anyway?

# Why a naive comparison fails

**Naive approach:** compare college GPA of scholarship recipients vs. non-recipients.

**Problem:** the two groups differ in *exactly the trait that determined the scholarship*: PSAT score.

- High-PSAT students tend to be more academically prepared
- They have better study habits, family resources, schools
- They were going to do better in college *regardless* of the scholarship

Classic selection bias. The naive comparison conflates the scholarship's effect with the effect of being a high-scoring student.

# Why our other methods don’t work here

**Can we use an RCT?** No. The scholarship isn’t randomly assigned. It’s awarded by rule.

**Can we use DiD?** No. There’s no clean “before” period for the scholarship — it’s awarded once, near the end of high school.

**Can we use panel methods?** Also no. Scholarship status doesn’t switch on and off for the same student over time.

**Can we use IV?** Maybe, but finding a valid instrument is hard.

We need a different approach.

# The RDD insight

Students who scored *exactly at the cutoff* are essentially indistinguishable from each other.

- A student who scored 1 point below the cutoff and one who scored 1 point above are, in expectation, identical
- Their academic ability, family background, motivation, and so on are statistically the same
- Yet one gets the scholarship and one doesn't

Right at the cutoff, we have something that looks like random assignment.

We can use this to identify the causal effect of the scholarship — but only for students *near the cutoff*.

# Where this works: "humans embed jumps into rules"

This trick generalizes far beyond scholarships. Anywhere a rule creates a sharp threshold, there's an RDD opportunity.

**Common rule-based cutoffs:**

- Test score thresholds (admissions, certifications, scholarships)
- Age cutoffs (drinking age, retirement age, voting age, Medicare)
- Income thresholds (tax brackets, benefit eligibility)
- Geographic boundaries (school districts, jurisdictions)
- Vote share thresholds (close elections)
- Environmental thresholds (monitoring/enforcement cutoffs)

Wherever bureaucratic rules create a discontinuity, RDD can exploit it.

# The Intuition

# The one-sentence intuition

*A rule assigns treatment based on whether a score crosses a cutoff.*

*People just below the cutoff are a good stand-in for what people just above would have looked like without treatment.*

*So any jump in the outcome at the cutoff is the treatment's effect.*

Everything else in this lecture is making this rigorous — and defending it against threats.

# Vocabulary

Some terms you'll see throughout RDD:

- Running variable (also: assignment variable, forcing variable, eligibility variable): the variable $X$ that determines treatment. In our example: PSAT score.
- Cutoff (or threshold) $c_0$: the value of $X$ at which treatment switches. In our example: the PSAT cutoff for scholarship eligibility.
- Treatment indicator $D$: whether the unit received treatment. In our example: $D_i = 1$ if student $i$ got the scholarship, 0 otherwise.
- Outcome $Y$: what we want to measure. In our example: college GPA.

# The continuity assumption (informally)

The core identifying idea:

> *In the absence of treatment, the relationship between $X$ and $Y$ would be smooth through the cutoff.*

**In our example:** if scholarships had no effect, college GPA would be a smooth function of PSAT score — no sudden jumps at the cutoff.

We can't directly observe this counterfactual (we don't see the no-scholarship outcome for students who got the scholarship). But we can make the case for it using institutional knowledge and indirect evidence.

**Cunningham's framing:** continuity should be your null hypothesis. *Nature does not make jumps.* A jump in the outcome demands an explanation. RDD argues the explanation is the treatment.

# "A turtle on a fencepost"

*If you see a turtle on a fencepost, you know he didn't get there by himself.*

A discontinuity in the outcome is the turtle on the fencepost.

Something *put it there*. RDD's job is to argue that the "something" was the treatment — not some confounder also changing at the cutoff.

This will be the recurring question throughout this lecture: how do we make sure the only thing changing at the cutoff is the treatment?

# Why this is essentially local randomization

Think about two students:

- Anna scored 1 point below the cutoff. No scholarship.
- Ben scored 1 point above the cutoff. Scholarship.

On any *observable* characteristic, Anna and Ben should look the same:

- Similar academic ability (1 point is noise)
- Similar family background, on average
- Similar test-day luck

The *only* systematic difference between them is the scholarship.

Right at the cutoff, treatment is essentially randomly assigned. This is what makes RDD work.

# The estimand: LATE at the cutoff

Because identification happens *only at the cutoff*, RDD estimates a Local Average Treatment Effect — the effect for units *at the cutoff*.

$$\delta_{RDD} = E[Y_i^1 - Y_i^0 \mid X_i = c_0]$$

**In our example:** we estimate the average effect of the scholarship *for students who scored right at the cutoff*.

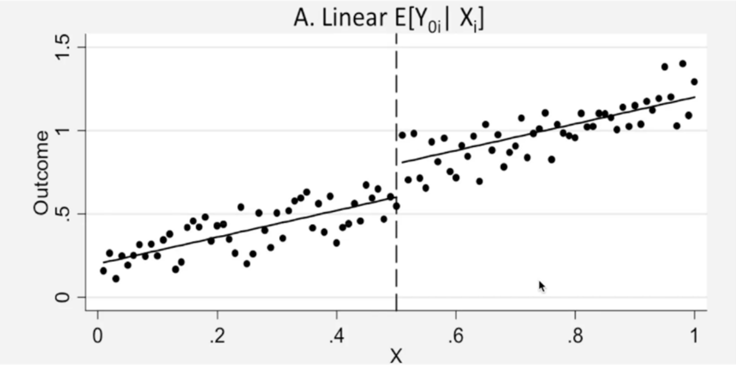
We do *not* estimate:

- The effect for students far above the cutoff
- The effect for students far below the cutoff
- The average effect across all students

This is a real limitation. We'll formalize this when we contrast ATE vs. LATE later.

# How Does RDD Work? Visual Treatment

# The best-case scenario: linear relationship



# Interpreting the linear case

**Setup of the figure:**

- X-axis: PSAT score (out of 100)
- Y-axis: college GPA (out of 1.5 — suspend disbelief)
- Threshold: $x = 0.5$

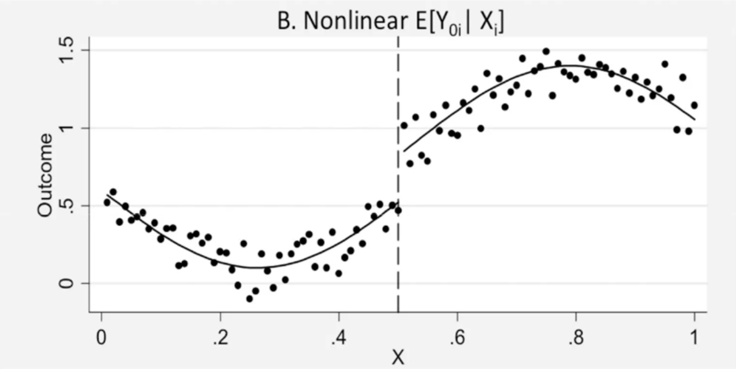
**What we see:**

- Higher PSAT scores associate with higher college GPAs (a positive relationship)
- The relationship between PSAT and GPA is linear on both sides of the cutoff
- There is a visible discrete jump at $x = 0.5$

The vertical gap at the cutoff is the estimated causal effect of the scholarship.

**Why this case is easy:** the relationship is linear, so we can fit straight lines on either side using all the data, then read off the gap at the cutoff

# A harder case: nonlinear relationship



# Interpreting the nonlinear case

**What we see:**

- The underlying relationship between PSAT and GPA is *nonlinear*
- The curve dips and rises on the left of the cutoff
- The curve rises and plateaus on the right
- There is still a visible discrete jump at the cutoff

**The challenge:** we can no longer fit straight lines. We need flexible curves on each side.

**The key point:** the jump itself is what identifies the treatment effect, regardless of the shape of the underlying relationship. As long as we correctly model the curves on each side, we can recover the discontinuity.

# What both cases share

In both the linear and nonlinear cases:

1. The relationship between $X$ and $Y$ is smooth on either side of the cutoff
2. There is a discrete vertical jump at the cutoff
3. The size of that jump is our estimate of the treatment effect

The shape of the underlying curve doesn't matter — what matters is the jump.

This is the visual essence of RDD. The next question: how do we measure that jump statistically?

# Estimation: Three Regression Specifications

# From pictures to regressions

We can measure the jump using regression. Three increasingly flexible specifications:

1. Pure linear: fit straight lines, allow a discrete intercept shift
2. Polynomial: fit a flexible curve, allow a discrete intercept shift
3. Different trends on each side: allow both intercept *and* slope to differ across the cutoff

Each is more flexible than the last. The right choice depends on what the data looks like.

# Specification 1: pure linear

$$y = \beta_0 + \beta_1 X + \beta_2 \cdot I(X > X_{threshold}) + e$$

**Where:**

- $X$: running variable (PSAT score)
- $I(\cdot)$: indicator function $= 1$ if argument true, 0 otherwise
- $\beta_1$: slope of the linear relationship
- $\beta_2$: the treatment effect — the jump at the cutoff

**Interpretation:** this fits one straight line through all the data, but allows the intercept to shift up or down at the cutoff by $\beta_2$.

**Best used when:** the relationship is genuinely linear on both sides (case A from our figures).

# Specification 2: polynomial

$$y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \beta_4 \cdot I(X > X_{threshold}) + e$$

**Where:**

- Higher-order terms $X^2, X^3$ allow for curvature
- $\beta_4$: the treatment effect — the jump at the cutoff

**Interpretation:** fit a flexible polynomial curve through the data, allow the curve to shift up or down at the cutoff.

**Best used when:** the relationship is curved (case B from our figures).

**Warning:** high-order polynomials (4th order and above) often *overfit* and put strange weight on observations far from the cutoff. Modern advice (Gelman & Imbens, 2019): stick to linear or quadratic.

# Specification 3: different trends on each side

First, recenter: $X' = X - X_0$ (where $X_0$ is the cutoff).

$$y = \beta_0 + \beta_1 X' + \beta_2 \cdot I(X > X_{threshold}) \cdot X' + \beta_3 \cdot I(X > X_{threshold}) + e$$

**Where:**

- $\beta_1$: slope on the *left* of the cutoff
- $\beta_1 + \beta_2$: slope on the *right* of the cutoff (because of the interaction)
- $\beta_3$: the treatment effect — the jump at the cutoff

**Interpretation:** fit two completely different lines on either side. The slopes can differ, and the intercept jumps at the cutoff.

**This is the most flexible of the three, and the modern default in serious applied work.**

# Why recentering matters

We define $X' = X - X_0$ so that $X' = 0$ at the cutoff.

**Why bother?**

- Without recentering, the intercept $\beta_0$ is the value of the regression at $X = 0$ (which may be meaningless)
- With recentering, $\beta_0$ is the value of the regression *at the cutoff* on the control side
- And $\beta_3$ reads directly as the jump *at the cutoff*

Recentering changes only the interpretation of the intercept, not the estimated treatment effect.

In practice, almost every RDD paper recenters.

# Bandwidth: how much data should we use?

A practical question: should we use *all* the data, or only observations near the cutoff?

**Wider bandwidth (use more data):**

- More observations $\Rightarrow$ less variance, tighter confidence intervals
- But: observations far from the cutoff influence the fit at the cutoff
- If the functional form is wrong, this introduces bias

**Narrower bandwidth (only data near the cutoff):**

- Less bias — linearity is a better approximation locally
- But: fewer observations $\Rightarrow$ more variance

This is the classic bias-variance trade-off. There's no free lunch.

## Modern practice: `rdrobust`

The bandwidth choice was historically arbitrary. Modern econometric tools automate it.

`rdrobust` **(Calonico, Cattaneo & Titiunik, 2014):**

- Picks an optimal bandwidth based on the data
- Allows different bandwidths on each side of the cutoff
- Provides bias-corrected, robust confidence intervals
- Available in R and Stata

**Practical workflow:**

1. Plot your data first — always
2. Run `rdrobust` with default settings as your baseline
3. Report results at a few alternative bandwidths to show robustness

This is what students should default to in their own projects.

# When Does RDD Fail?

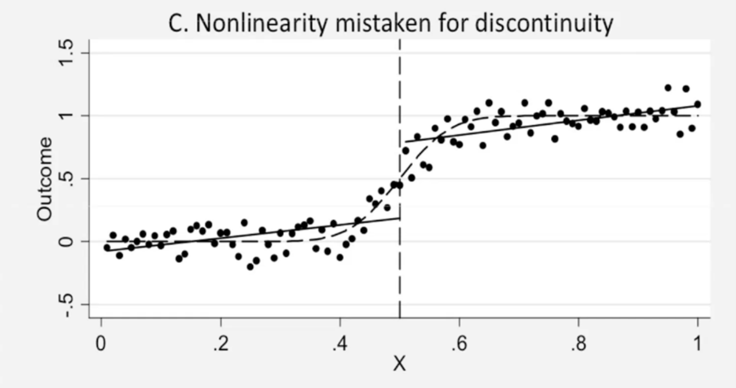
# Four ways RDD can fail

We've seen the visual logic and the regression. Now: when does this go wrong?

1. The relationship is curved, but we fit straight lines — spurious jumps
2. Another program uses the same threshold — confounded effects
3. The underlying relationship is jumpy everywhere — the cutoff jump isn't special
4. Individuals manipulate $X$ to push themselves over the threshold — sorting bias

We'll go through each.

# Failure 1: Curvature mistaken for discontinuity



# The kink vs. the discrete jump

**What's happening in the figure:**

- The true relationship is smooth but *curved* (the dashed line) — an S-shape
- No discrete jump exists at the cutoff
- But fitting *straight lines* on each side (the solid lines) creates an apparent jump

**Crucially: this is a kink, not a discontinuity.**

- A kink: the slope changes, but the function is still continuous (no vertical jump)
- A discrete jump: the function value itself changes abruptly at the cutoff

RDD identifies discrete jumps, not kinks. If you fit linear models to a curved relationship, you'll "find" a jump that isn't there.



**Defense:** fit flexible curves. Plot your data. If the loess fit (a smooth

# Failure 2: another program at the same threshold

What if *something else* also changes at the cutoff?

**Example: Medicare at age 65.**

- Medicare eligibility turns on sharply at age 65
- But age 65 is *also* the traditional retirement age
- And many other policies, social norms, and life events shift at 65

If you find a jump in healthcare utilization at 65, is it from Medicare? Or from retirement? Or from something else?

The cutoff is endogenous — multiple things change at the same place.

# Defending against endogenous cutoffs

**Defense:** test for jumps in other outcomes or covariates that should *not* be affected by the treatment.

**Example (Card, Dobkin & Maestas, 2008):**

- Looking at Medicare effects at 65
- Worried about retirement at 65
- Checked: is there a discontinuity in *employment* at 65?
- Answer: no. Employment trends smoothly through 65 in their data.
- Conclusion: retirement isn't driving the jump — it must be Medicare

This is a placebo test: look for effects where you shouldn't find any. Finding effects in the wrong places is a red flag.

# Failure 3: jumpy underlying relationships

What if the relationship between $X$ and $Y$ has *many* jumps, not just at the cutoff?

**Example:** suppose GPAs as a function of PSAT score have natural jumps at every multiple of 10 (because PSAT scores cluster on round numbers).

Then the "jump" at the cutoff isn't special — it's just one of many.

**Defense: placebo cutoffs.**

- Pretend the cutoff is at some other value where no policy actually changes
- Run the same RDD specification
- You should find *no* effect
- If you find effects at fake cutoffs, the underlying data is too jumpy for RDD

# Failure 4: manipulation at the threshold

**The biggest threat to RDD.** If people can adjust their $X$ to land on the favorable side of the cutoff, treatment is no longer "random" near the cutoff.

**When is manipulation possible?**

1. The assignment rule is *known* in advance
2. The treatment is *desirable*
3. Agents have *time and ability* to adjust

**Concrete examples:**

- PSAT cutoff: could a student retake the test? Are graders flexible at the cutoff?
- Low birthweight 1500g threshold: do hospitals round down to qualify for treatment?
- Income cutoffs for benefits: do people underreport income just above the cutoff?

# Detecting manipulation: the density test

**Intuition (Cunningham's "two rooms" story):**

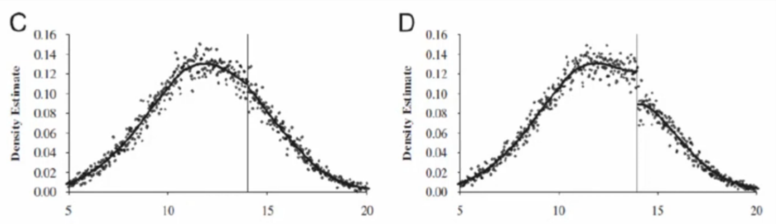
- Room A: gets a life-saving treatment
- Room B: gets nothing
- Everyone knows which room is which

**What do people in Room B do?** They walk to Room A if they can.

If sorting succeeded, an outsider would see Room A crowded, Room B empty.

**The test:** examine the distribution of the running variable near the cutoff. If there's an excess of observations just above the cutoff (or a deficit just below), that's evidence of manipulation.

# Visualizing sorting (or lack of it)



# Reading the density plots

**Left panel (C):**

- The density of the running variable trends smoothly through the cutoff
- No suspicious clustering on either side
- Consistent with no manipulation

**Right panel (D):**

- There is a sharp jump in density at the cutoff
- Excess mass on the treated side
- Strong evidence of manipulation

If you see panel D in your data, RDD is in serious trouble.

**The formal version:** the McCrary (2008) density test (in R: `rddensity`). It quantifies whether the density discontinuity at the cutoff is statistically significant.

# Important caveat about the density test

**Passing the density test does not prove there was no manipulation.**

Why? Because:

- Manipulation might be *two-sided* (some people move up, some move down) — this leaves the density smooth even though sorting is happening
- The test is underpowered with small samples
- Manipulation might be at a different level than your test detects

Absence of evidence is not evidence of absence.

The density test is a useful check, but it should be combined with:

- Institutional knowledge (could manipulation realistically occur here?)
- Covariate balance tests (do pretreatment characteristics jump at the cutoff?)
- Placebo tests at other thresholds

# Sharp vs. Fuzzy RDD

# Sharp vs. fuzzy: a preview

So far we've assumed that crossing the cutoff *guarantees* treatment.

**Sharp RDD:** crossing the cutoff determines treatment.

$$D_i = \begin{cases} 1 & \text{if } X_i \geq c_0 \\ 0 & \text{if } X_i < c_0 \end{cases}$$

**Fuzzy RDD:** crossing the cutoff increases the probability of treatment, but doesn't guarantee it.

$$\Pr(D_i = 1 \mid X_i \geq c_0) > \Pr(D_i = 1 \mid X_i < c_0)$$

In a fuzzy RDD, the jump is in the *probability of treatment*, not in treatment itself.

# Why does fuzzy RDD happen?

**In our scholarship example:** crossing the PSAT cutoff makes you *eligible* for the scholarship. But:

- Some eligible students don't apply or accept
- Some students below the cutoff might receive the scholarship through other criteria
- Some students take the scholarship and reject it later

So in practice, the relationship between PSAT $\geq$ cutoff and "receives scholarship" isn't 0-to-1 — it might be 0.05 to 0.85.

**Many real-world RDDs are fuzzy:**

- College admission thresholds (eligibility $\neq$ enrollment)
- Class size rules (don't perfectly determine actual class size)
- Benefit cutoffs (eligibility $\neq$ take-up)

# The identifying assumption is the same

**For both sharp and fuzzy RDD, the identifying assumption is continuity.**

What *differs* is what jumps at the cutoff:

- Sharp: treatment $D$ jumps from 0 to 1
- Fuzzy: probability of treatment jumps by some amount less than 1

What *stays the same:* potential outcomes $Y^0$ and $Y^1$ must trend smoothly through the cutoff in both cases.

# Fuzzy RDD as an IV problem

The technical fix for fuzzy RDD uses instrumental variables.

**Idea:** use "crossed the cutoff" as an instrument for "received treatment."

**Why this is a brief preview only:** we haven't covered IV yet. We'll formalize this in a later lecture.

**The intuition:**

- The cutoff strongly predicts who gets treated (the "first stage")
- The cutoff doesn't directly affect outcomes *except* through treatment (the "exclusion restriction")
- So we can use the cutoff to isolate the part of treatment that's exogenous

**When you get to IV later in the course, you'll see fuzzy RDD as a clean application of it.**

# ATE vs. LATE

# Two kinds of treatment effects

A crucial distinction that's been lurking in the background:

**Average Treatment Effect (ATE):**

$$\text{ATE} = E[Y_i^1 - Y_i^0]$$

The average effect of treatment *across the entire population*.

**Local Average Treatment Effect (LATE):**

$$\text{LATE} = E[Y_i^1 - Y_i^0 \mid \text{some subgroup}]$$

The average effect of treatment *for a specific subgroup*.

RDD estimates a LATE, not an ATE.

# What's the "local" subgroup in RDD?

For sharp RDD: the subgroup is units at the cutoff.

$$\text{LATE}_{RDD} = E[Y_i^1 - Y_i^0 \mid X_i = c_0]$$

**In our example:** the average effect of the scholarship *for students who scored right at the cutoff*.

For fuzzy RDD: it's even more local — the effect for compliers at the cutoff (those whose treatment status flips because of the cutoff).

# Why ATE and LATE can differ

Treatment effects often *vary across the population*.

**In our scholarship example:**

- Students far above the cutoff probably don't need the scholarship's financial component — they have other options
- Students far below the cutoff might benefit enormously from the financial support — if they could get it
- Students at the cutoff are somewhere in between

The effect we estimate (at the cutoff) might not match the effect for either group far from it.

LATE = the effect for the marginal student. It's policy-relevant, but it's not the average effect across all students.

# ATE vs. LATE: strengths and limitations

| **ATE** | **LATE** |
|---|---|
| Average effect across the entire population | Effect only for a specific subgroup |
| What you'd want for general policy decisions | What you get from RCTs (when noncompliance exists) and from RDD |
| Requires unbiased estimation across the population — usually demanding | Requires only local unbiasedness — much weaker requirement |
| Strong external validity | Limited external validity |

Trade-off: ATE answers a more general question but requires stronger assumptions. LATE answers a narrower question but with weaker assumptions.

# Identifying Assumptions

# The continuity assumption (formally)

Continuity assumption: In the absence of treatment, the conditional expectations of potential outcomes are continuous at the cutoff.

$$\lim_{X \to c_0^-} E[Y^0 \mid X] = \lim_{X \to c_0^+} E[Y^0 \mid X]$$
$$\lim_{X \to c_0^-} E[Y^1 \mid X] = \lim_{X \to c_0^+} E[Y^1 \mid X]$$

**In plain English:** no other variable that affects $Y$ also changes discontinuously at the cutoff.

**What it rules out:**

- Other policies or programs activated at the same cutoff
- Strategic sorting by units
- Anything else that creates a jump at $c_0$

# Why continuity is untestable

We never observe both potential outcomes for the same unit.

For units *above* the cutoff: we see $Y^1$ but not $Y^0$.
For units *below* the cutoff: we see $Y^0$ but not $Y^1$.

So we can *never* directly check whether $E[Y^0 \mid X]$ is continuous at the cutoff — we've only ever observed it on one side.

Continuity is an assumption, not a finding.

**What we can do:** build a circumstantial case using:

1. Institutional knowledge (why would $a_i$ jump at $c_0$? Is there a competing program?)
2. Covariate balance (do pretreatment variables show jumps?)
3. Density tests (is there sorting?)
4. Placebo tests (do we find effects at fake cutoffs?)

# Validation Strategy 1: covariates against the running variable

**Idea:** pretreatment characteristics — things that physically *cannot* be caused by the treatment — should not jump at the cutoff.

**In our scholarship example:**

- Race, sex, family income (measured before the test) shouldn't jump at the cutoff
- Parental education shouldn't jump at the cutoff
- Past academic performance (early high school grades) shouldn't jump at the cutoff

**The procedure:**

1. Pick a pretreatment covariate
2. Plot it against the running variable
3. Look at the cutoff: is there a visible discontinuity?
4. If yes: continuity is in trouble. If no: reassurance (but not proof).

# Validation Strategy 2: fitted values from covariates

**Idea:** build a model that predicts $Y$ using only pretreatment covariates — not treatment. Then check whether the predictions jump at the cutoff.

**The procedure:**

1. Regress $Y$ on a set of pretreatment covariates: $Y = \gamma_0 + \gamma_1 W_1 + \gamma_2 W_2 + \cdots + \varepsilon$
2. Get fitted values: $\hat{Y}$
3. Plot $\hat{Y}$ against the running variable $X$
4. Look for a discontinuity at the cutoff

**What we want to see:** smooth $\hat{Y}$ through the cutoff. If $\hat{Y}$ (which uses no treatment information) jumps at the cutoff, then something other than treatment is jumping.

Smoothness of $\hat{Y}$ at the cutoff = reassurance that the discontinuity in $Y$ is genuinely due to treatment.

# Validation Strategy 3: RDD on covariates

**Idea:** run the same RDD specification using a *pretreatment covariate* as the outcome. The coefficient on the treatment indicator should be *insignificant*.

**The procedure:**

1. Pick a pretreatment covariate $W$ (race, family income, parental education)
2. Estimate the RDD-style equation:

$$W = \beta_0 + \beta_1 X + \beta_2 \cdot I(X > X_{threshold}) + e$$

3. Examine $\hat{\beta}_2$ — the "treatment effect" on the pretreatment covariate
4. It should be *statistically insignificant*

**What we want to see:** $\hat{\beta}_2 \approx 0$ for every pretreatment covariate.

A significant "effect" on something that couldn't possibly be caused by treatment is a strong signal that the design is broken.

# The no-sorting assumption

No sorting at the cutoff: individuals cannot strategically manipulate $X$ to land on the favorable side of the cutoff.

This is implied by continuity — if people sort across the cutoff based on their potential outcomes, then $E[Y^0 \mid X]$ will be discontinuous.

**When this assumption fails:**

- Manipulators end up on the treated side
- Manipulators have different $E[Y^0]$ than non-manipulators
- The cutoff jump partly reflects sorting, not treatment

**Defense:** the density test we covered earlier. Plot the distribution of $X$ near the cutoff; look for clustering on one side.

## Summary: how to defend an RDD

A credible RDD paper does *all* of the following:

1. Plot the data. Show the raw scatter, the binned means, and the fits on each side of the cutoff.
2. Density test. Show the distribution of the running variable is smooth at the cutoff.
3. Covariate balance. Show pretreatment variables don't jump at the cutoff.
4. Placebo cutoffs. Show no "effects" appear at fake cutoffs.
5. Robustness across bandwidths. Show estimates are stable as you vary how much data you include.
6. Institutional knowledge. Explain *why* you believe continuity holds in this specific setting.

# Where RDD is Used

# Where RDD is used in practice

RDD is now one of the most popular causal inference designs. From a handful of papers in the 1980s to over 5,600 by 2019.

Why? Two reasons:

- The credibility revolution: economists value transparent, defensible designs
- Administrative data has become widely available — and it usually encodes the bureaucratic cutoffs RDD exploits

Wherever organizations face scarcity and ration by some scored cutoff, there's a potential RDD.

# Examples: public policy and government

RDD is heavily used in evaluating government policy:

- Education policy: effects of school admission cutoffs, class size rules, scholarship thresholds
- Healthcare policy: effects of Medicare eligibility, drug coverage thresholds
- Welfare policy: effects of income-based eligibility cutoffs for benefits programs
- Crime policy: effects of legal drinking age, age cutoffs for adult vs. juvenile sentencing
- Tax policy: effects of income thresholds for tax brackets, deductions, credits

Government agencies frequently *create* these cutoffs as part of policy design — so the data needed for RDD often exists in administrative records.

## Examples: environment and natural resources

- Deforestation enforcement: environmental agencies often target enforcement based on deforestation rates crossing a threshold. RDD can identify whether enforcement reduces deforestation.
- Air quality regulation: pollution levels above a threshold trigger regulatory action. RDD can measure the effect of regulation.
- Geographic boundaries: environmental protections that switch on at park boundaries, river boundaries, etc.
- Conservation programs: eligibility cutoffs for payments to landowners.

**Example application: tropical deforestation.** Countries (including Brazil) have set deforestation rate thresholds that trigger increased monitoring or enforcement. RDD has been used to estimate the effect of these enforcement triggers on subsequent deforestation rates.

# Examples: private sector and business

RDD shows up regularly in industry too:

- Credit scoring: effects of credit score cutoffs on loan approval, default rates, downstream consumer behavior
- Customer programs: effects of loyalty program tiers (Silver/Gold/Platinum) based on spending cutoffs
- Marketing: effects of pricing thresholds (free shipping above $50, discount at quantity thresholds)
- Hiring: effects of test score or experience cutoffs in employee screening
- Online platforms: effects of features unlocked at certain user thresholds (review count, follower count)

Tech companies in particular use these techniques to measure causal effects of product changes when randomization isn't feasible.

# Why this is everywhere

> *Firms and government agencies are unknowingly sitting atop a mountain of potential RDD projects.*
>
> — Cunningham, *Causal Inference: The Mixtape*

Any organization facing scarcity rations its treatment somehow. A cutoff on a continuous score is one of the most common rationing devices — because it's transparent, auditable, and seems fair.

For applied economists and data analysts: this means RDD opportunities exist wherever you look. The hard part isn't finding the design — it's defending continuity in your specific setting.

# Wrap-up

# Advantages of RDD

- Credibly defeats selection bias. Treatment near the cutoff is determined by the rule, not by unit characteristics.
- Transparent assumptions. Continuity is easier to reason about (with institutional knowledge) than assumptions behind matching or naive regression.
- Visually persuasive. A skeptical reader can almost *see* whether the design works.
- Falsifiable. Density tests, covariate balance, placebo cutoffs — the design comes with built-in checks of its own credibility.
- Opportunities everywhere. Rule-based cutoffs are everywhere in policy and business.

# Limitations of RDD

- LATE only. Estimates apply only to units at the cutoff. If effects vary across the population, RDD won't tell you about that variation.
- Continuity is untestable. The whole design rests on an assumption you cannot directly verify.
- Data-hungry. You need many observations near the cutoff. Small samples make RDD unreliable.
- Functional form sensitivity. Curves mistaken for jumps; high-order polynomials overfit; estimates can drift across specifications.
- Vulnerable to manipulation. If units can sort across the cutoff, the design breaks.
- Vulnerable to endogenous cutoffs. If multiple things change at the same threshold, you confound their effects.

# Recap

1. RDD solves a problem the other methods can't: causal identification when treatment is assigned by a rule.
2. Identification comes from local randomization at the cutoff — units just above and just below are essentially indistinguishable except for treatment.
3. Estimation: fit flexible functions on each side, measure the vertical gap at the cutoff. Three regression specifications, from simple linear to fully flexible.
4. Threats: nonlinearity mistaken for discontinuity, endogenous cutoffs, jumpy data, manipulation/sorting.
5. Identifying assumption: continuity. Untestable directly. Defended via institutional knowledge, density tests, covariate balance, placebo cutoffs.
6. Estimand is a LATE — the effect at the cutoff, not the average across the population.

# Next time

**Instrumental variables.**

We've now seen four ways to handle selection bias:

- Random assignment (RCTs)
- Within-group comparisons (DiD, panel methods)
- Local randomization (RDD)

Next: a fifth approach. When you can't randomize and don't have a sharp cutoff, you can sometimes find an external variable that influences treatment but not outcomes directly. That variable is an instrument.

This will also let us formalize fuzzy RDD.

**Reading:** Cunningham, *Causal Inference: The Mixtape*, Chapter 6 (RDD); Angrist & Pischke, *Mostly Harmless Econometrics*, Ch. 6.

## References and further reading


- Cunningham, S. (2021). *Causal Inference: The Mixtape.* Yale University Press. (Ch. 6)
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